

MACHINE LEARNING

What is

LEARNING?

Learning** is the acquisition of knowledge or skills through **study, experience, or being taught.

The background of the slide is a dark, textured surface covered with various mathematical formulas in a light green or yellowish font. These formulas are slightly blurred and overlap each other, creating a sense of depth and complexity. Some visible formulas include trigonometric identities, algebraic expressions, and calculus notations.

What is
MACHINE LEARNING?

***Machine Learning** is a type of
artificial intelligence that allows
machines to learn from **data** without
being explicitly programmed.*

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Importance of
MACHINE LEARNING?

Importance of machine learning

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graph LR; A[Importance of machine learning] --- B[Uncover patterns in large dataset]; A --- C[Automate decision making]; A --- D[Other applications across industries];
```

Uncover patterns in large dataset

Automate decision making

Other applications across industries

☐ Supervised
learning

☐ Unsupervised
learning

☐ Reinforcement
learning

types of
MACHINE LEARNING

❑ Supervised
learning

❑ Unsupervised
learning

❑ Reinforcement
learning

*This is a type of machine learning in which a model is trained to learn from **labeled data***

*It's **goal** is to predict outcomes for new data*

😊 Matthew 25:

❑ Supervised learning

❑ Unsupervised learning

❑ Reinforcement learning

*This is a type of machine learning in which a model is trained to identify patterns **unlabeled data***

*It's **goal** is to get insight from large volume of data*

😊 genesis 11:

☐ Supervised learning

☐ Unsupervised learning

☐ Reinforcement learning

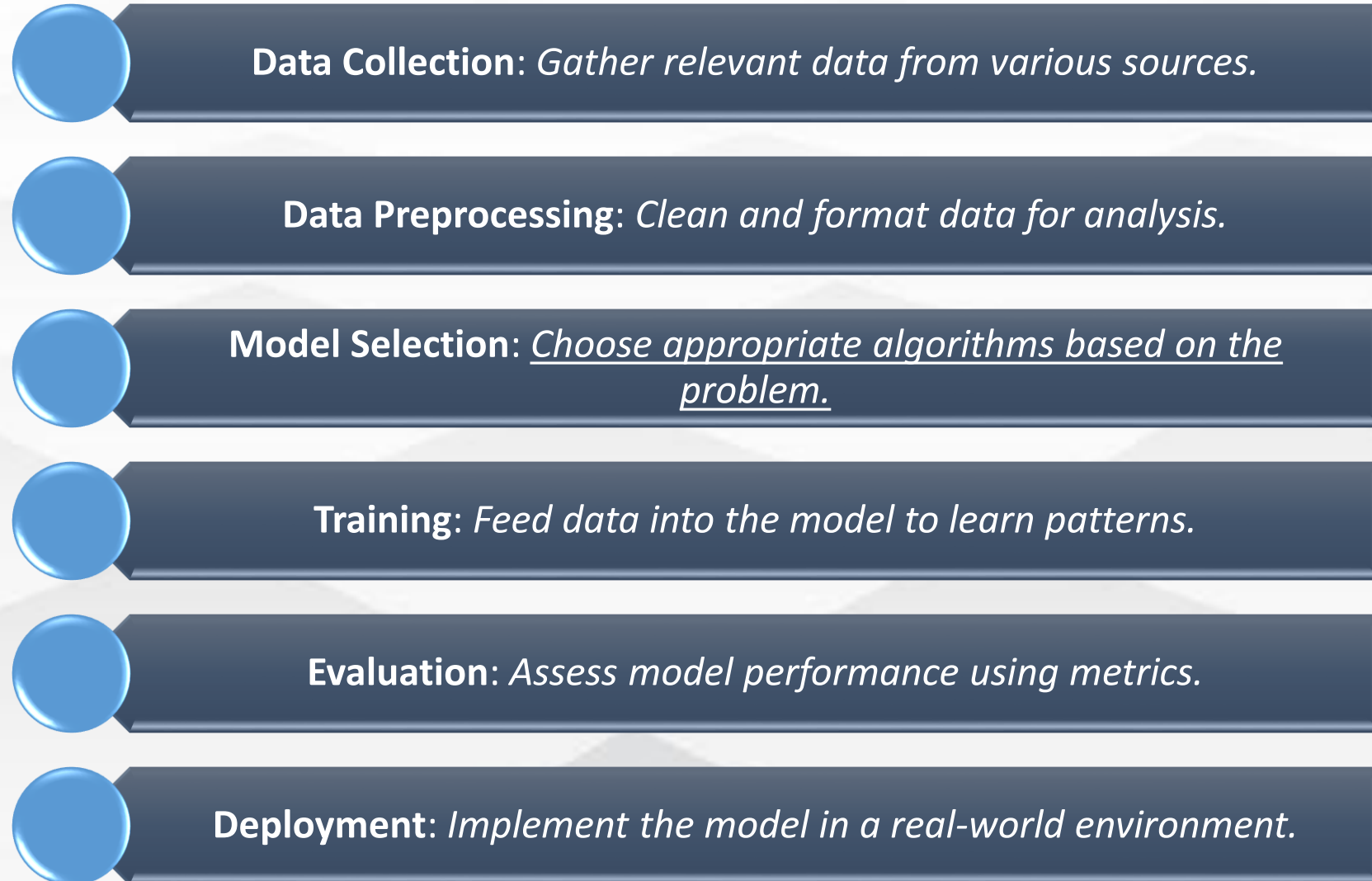
*This is a type of machine learning in which a model is trained to learn by **receiving reward or penalty** for actions*

*It's **goal** is to **minimize penalties** and **maximize rewards***



MACHINE LEARNING

workflow



MACHINE LEARNING

problems

Classification : Predicting discrete output values using input variables .

Clustering : Creating subgroups based of common shared features .

Regression : Predicting continues output using input variables
:

For casting : This is a time – indexed prediction
.

Recommendation : making an offer based on receiver's preference.

Assign the above ML problems
into either supervised or
unsupervised ML problem

Linear Regression
Logistic Regression
Decision Tree
Random forest
Support Vector Machine
K-Nearest Neighbors
Naïve Bayes

K-Means Clustering
Hierarchical Clustering
Principal Component Analysis
DBSCAN

Q-Learning
Deep Q-learning
Policy Gradient Method

MACHINE LEARNING

Algorithms

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Correlation and linear Regression

Correlations: they describe the strength and direction of relationship between Two variables X and Y

This is done using the pearson's correlation coefficient (r)

$$r = \frac{cov(X, Y)}{\sigma_x \sigma_y}$$

$$cov(X, Y) = \frac{\sum_{i=1}^N (x - \bar{x})(y - \bar{y})}{N - 1}$$

$$\sigma_x^2 = \frac{\sum_{i=1}^N (x - \bar{x})^2}{N - 1}$$

Positive correlation : as one increase the other increase
Negative correlation : as on increase the other decreases
No correlation(r = 0) : no linear relationship

r ranges from -1 to 1

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Correlation and linear Regression

Problem with correlations:

- It is bidirectional i.e there is NO dependent and independent variable (cannot be used for prediction)
- It does not tell us how a change in one variable will affect a change in the other variable

what is Linear Regression

Linear regression is a statistical technique used to estimate the relationship between variables, typically to predict one based on the other.

Using the equation of a straight line

$$y = mx + c$$

y : dependent (target variable)

x : independent variable

m : slope(in this context a weight)

c : intercept

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Linear Regression Model

$$y_i = \beta_0 + \beta_i x_i + \varepsilon$$

y_i : dependent (target variable)

x_i : independent variable

β_i : weights(*telling us how a change in their host feature will affect a change in y_i*)

β_0 : intercept

ε : error term

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Assumptions for linear Regression

Linearity: the relationship between the target and features is linear ($y = mx + c$)

No Autocorrelation between errors : error should be independent of each other (detect using Durbin-Watson test)

No collinearity: independent variable shouldn't highly correlated

Normality of errors: errors should be normally distributed

Homoscedasticity: variance of errors should be constant across All levels of the independent variables

No outliers: as they can distort the regression line (detect using boxplot
cook's distance and leverage plot)

$$y_i = \beta_0 + \sum_{i=1}^n \beta_i x_i + \varepsilon$$

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linear Regression implementation In python